UNITED INTERNATIONAL UNIVERSITY (UIU)

Dept. of Computer Science & Engineering

Class Test 4 (Set C)

Course No: CSE 4325 Course Title: Microprocessors and Microcontrollers

Time: 15 minutes **Marks: 20**

Name		**Section**		**ID**	

1. Suppose there are **nine instructions** loaded into the Instruction Memory (within the Code Segment of 8086). The BIU of the 8086 will fetch these instructions for execution by the EU.

   When the one instruction is being executed, three instructions are fetched and saved in the instruction queue. Draw the corresponding BIU and EU cycles for the following cases.

   a. The 4th instruction is a 'MOV <address>', the 6th is a 'JUMP <8th instruction address>' and the 7th instruction is a 'MOV <address>' instruction. (10)
   b. The 1st instruction is a 'MOV <address>', the 5th is a 'JUMP <7th instruction address>' and the 7th instruction is a 'JUMP <4th instruction address>' instruction. (10)